

Speak Like A Mathematician:

CPCTC:

corresponding parts:

Using CPCTC

**CPCTC stands for _____ of
_____ are
_____.**

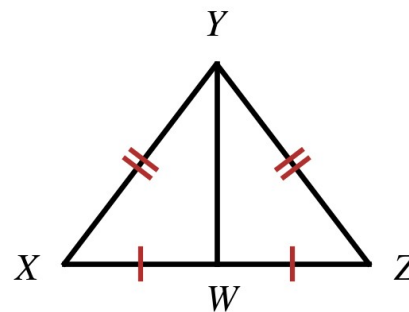
**Order matters: FIRST prove the two triangles
_____ with SSS, SAS, ASA, AAS, or HL. THEN
use CPCTC to conclude that any other pair of parts is
congruent.**

**CPCTC is never the reason that makes two triangles
congruent — it is what you get _____ they are.**

Example 1 - A Two-Step Plan

Before writing a CPCTC proof, plan it in two sentences.

- To prove $\angle X \cong \angle Z$ in the diagram, which two triangles would you use?
- Which criterion will prove those triangles congruent, and why?
- What is the very last line of the proof?

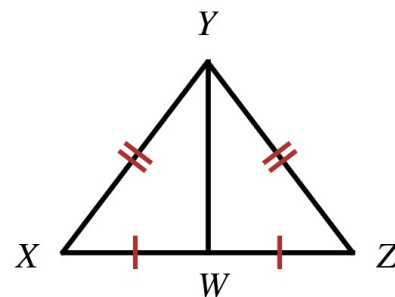


Example 2 - Prove Two Angles Congruent

Given: $\overline{YW}$ bisects $\overline{XZ}$; $\overline{XY} \cong \overline{YZ}$

Prove: $\angle X \cong \angle Z$

Statements	Reasons
1. $\overline{YW}$ bisects $\overline{XZ}$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.

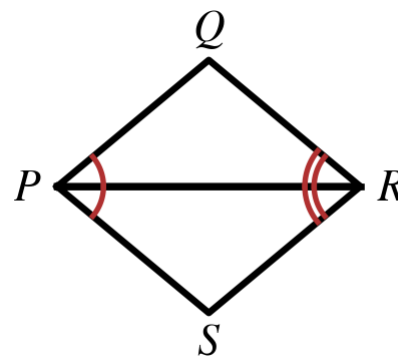


Example 3 - Prove Two Sides Congruent

Given: $\overline{PR}$ bisects $\angle QPS$ and $\angle QRS$

Prove: $\overline{PQ} \cong \overline{PS}$

Statements	Reasons
1. $\overline{PR}$ bisects $\angle QPS$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.

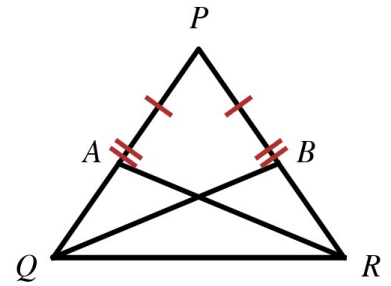


Example 4 - A Proof Inside an Isosceles Triangle

Given: Isosceles $\triangle PQR$ with base $\overline{QR}$; $\overline{PA} \cong \overline{PB}$

Prove: $\overline{AR} \cong \overline{BQ}$

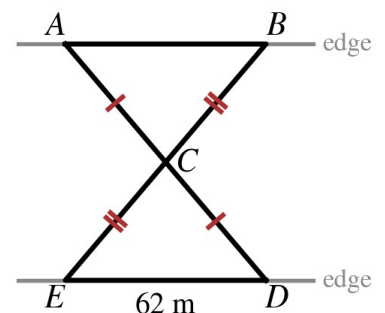
Statements	Reasons
1. $\triangle PQR$ is isosceles with base $\overline{QR}$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.



Example 5 - Measuring Across a Ravine

A and B are on the edges of a ravine with parallel sides. A surveyor marks C, then measures out to D and E so that $\overline{AC} \cong \overline{CD}$ and $\overline{BC} \cong \overline{CE}$. She measures $ED = 62$ m.

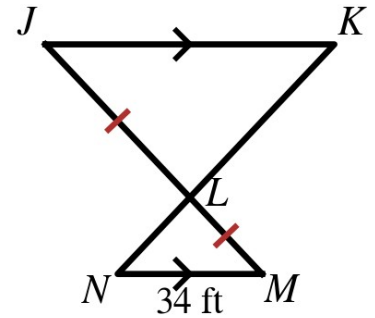
- Which two triangles are congruent, and why?
- Find AB. What reason justifies it?
- Why is it useful to measure ED instead of AB directly?



Example 6 - Measuring Across a Pond

A landscape architect needs the distance JK across a pond. She sets up the triangles at the right so that $\overline{JK} \parallel \overline{NM}$ and L is the midpoint of $\overline{JM}$. She measures $NM = 34$ ft.

- Name a pair of congruent angles created by the parallel sides. Give the reason.
- Name another pair of congruent angles at L . Give the reason.
- Prove the triangles congruent. Which criterion?
- Find JK and justify it.



Example 7 - Is There Enough Information?

Decide whether each plan works. Explain.

- "I showed two pairs of sides congruent, so I can use CPCTC."
- "The two triangles are congruent, so every pair of corresponding parts is congruent."